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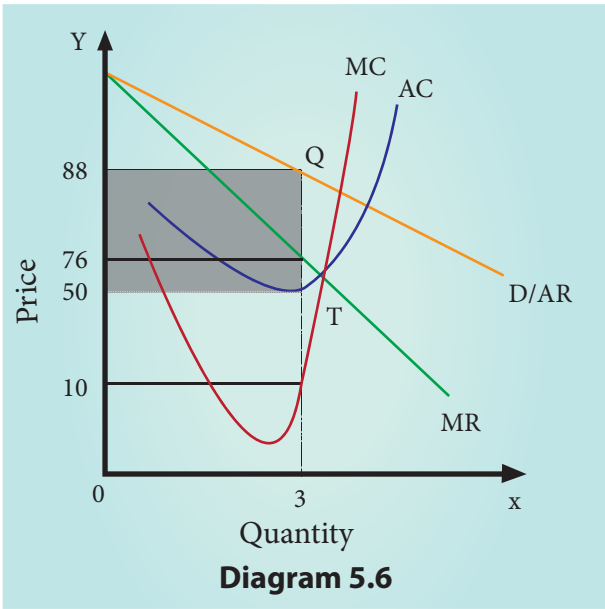


Diagram 5.6

as more units of output are sold, the MR lies below the AR curve (MR<AR).

The monopolist will continue to sell his product as long as his MR>MC.

He attains equilibrium at the level of output when its MC is equal to MR. Beyond this point, the producer will experience loss and hence will stop selling.

Let us take the following hypothetical example of Total Revenue Function and Total cost function.

TR=100Q-4Q2 and
TC=Q3 - 18Q2 + 91Q +12.

Therefore AR= 100 - 4Q;
MR=100 - 8Q;
AC= Q2- 18Q + 91 + 12/Q;

MC= $3Q^2$ - 36Q + 91;

When Q=3,

AR= 100 – 4(3) = 88,

AC= 3^2-18(3) +91+12/3=
9-54+91+4=50;

MR=100-8(3) =76;

MC= 3(3)2- 36(3) +91=27-108+
91 = 10

From this diagram, till he sells 3 units output, MR is greater than MC, and when he exceeds this output level, MR is less than MC. The monopoly firm will be in equilibrium at the level of output where MR is equal to MC. The price is 88.

To checkup how much profit the monopolist is making at the equilibrium output, the average revenue curves and the average cost curves are used. At equilibrium level of output is 3; the average revenue is 88 and the average cost is 50. Therefore (88-50 =38) is the profit per unit.

Total profit = (Average Revenue – Average Cost) X Total output

= (88 – 50) X3

= 38X3=114.

5.7.4 Price Discrimination under monopoly

A discriminating monopoly is a single entity that charges different prices for different consumers. Higher price will be charged for price inelastic consumers and vice versa

Types of Price Discrimination

There are three types of price discrimination.

(i) ***Personal*** – Different prices are charged for different individuals (for example, the railways give tickets at concessional rate to the *'senior citizens'* for the same journey).

(ii) ***Geographical*** - Different prices are charged at different places for the same product (for example, a book sold within India at a price is sold in a foreign country at lower price). On